

Lecture Notes I

Course: Functional Analysis

TA: Morales C. Ronny

Topic: The Lebesgue Outer Measure

Theorem 1.1. Suppose A is countable. If A is finite, say $A = \{a_k\}_{k=1}^n$, it is enough to choose the sequence

$$I_k = \left(a_k - \frac{\varepsilon}{2n}, a_k + \frac{\varepsilon}{2n}\right), \quad k = 1, \dots, n,$$

and thus $m^*(A) \leq \varepsilon$, and since $\varepsilon > 0$ was arbitrary, the conclusion follows. For the denumerable case we have that there is a bijection of A onto $\mathbb{N}$, say $\varphi : A \rightarrow \mathbb{N}$, so that we can define $r_i = \varphi^{-1}(i)$ and hence $A = \{r_n\}$ so that we define the sequence

$$I_k = \left(r_k - \frac{\varepsilon}{2^{k+1}}, r_k + \frac{\varepsilon}{2^{k+1}}\right),$$

and thus

$$\sum_{\mathbb{N}} \ell(I_n) = \sum_{\mathbb{N}} \frac{\varepsilon}{2^k} = \varepsilon,$$

so we conclude that $m^*(A) \leq \varepsilon$, and again, since $\varepsilon > 0$ was arbitrary, we conclude that $m^*(A) = 0$.

Theorem 1.2.

- i) Suppose $A \subseteq B$ and let (I_n) be a countable open cover of B , then it is also an open cover for A since

$$\bigcup_{n \in \mathbb{N}} I_n \supseteq B \supseteq A,$$

then the class $\mathcal{C}_A$ of all countable open covers of A contains the class $\mathcal{C}_B$ (defined analogously) and thus

$$\left\{ \sum_{n \in \mathbb{N}} \ell(I_n) : (I_n) \in \mathcal{C}_A \right\} \supseteq \left\{ \sum_{n \in \mathbb{N}} \ell(I_n) : (I_n) \in \mathcal{C}_B \right\},$$

whence $m^*(A) \leq m^*(B)$.

- ii) Suppose (I_n) is a countable open cover of A , then clearly $(I_n + t)$ is a countable open cover of $A + t$ and hence $\mathcal{C}$

$$m^*(t + A) \leq \sum_{n \in \mathbb{N}} \ell(I_n + t) = \sum_{n \in \mathbb{N}} \ell(I_n),$$

which on taking the infimum over $\mathcal{C}$ we get $m^*(A + t) \leq m^*(A)$. Conversely, note that $A = -t + (A + t)$

$$m^*(A) = m^*((A + t) + (-t))$$

and by letting $D = A + t$ this is just $m^*(D + (-t))$ which by the previous inequality satisfies $m^*(D + (-t)) \leq m^*(D) = m^*(A + t)$ and thus $m^*(A) \leq m^*(A + t)$.

- iii) If $|A_k| = \infty$ for some k , the conclusion readily follows, so suppose that $|A_k| < \infty$ for all $k \in \mathbb{N}$, then let (I_n^k) be a countable open cover of A_k for which

$$\sum_{n \in \mathbb{N}} \ell(I_n^k) \leq \frac{\varepsilon}{2^k} + m^*(A_k) \implies \sum_{k \in \mathbb{N}} \sum_{n \in \mathbb{N}} \ell(I_n^k) \leq \varepsilon + \sum_{k \in \mathbb{N}} m^*(A_k), \quad (*)$$

the double sum is written as

$$\sum_{(k,n) \in \mathbb{N}^2} \ell(I_n^k) = \sum_{\mu \in \mathbb{N}} \ell(I_\mu)$$

where in the last equality we used the (known to exist) bijection $\varphi : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ and wrote $\mu = \varphi(k, n)$. Given that the sequence (I_μ) contains each A_k , we can conclude that (I_μ) is a countable open cover and given last inequality in $(*)$, we also deduce that

$$m^* \left(\bigcup_{k \in \mathbb{N}} A_k \right) \leq \varepsilon + \sum_{k \in \mathbb{N}} m^*(A_k),$$

therefore, the required result follows from the arbitrariness of ε .

Theorem 1.4. Suppose K is closed and bounded and let $\mathcal{C}$ be an open cover of K .

Case 1) Suppose $K = [a, b]$ for some $a < b$. Define

$$D = \{d \in [a, b] : [a, d] \text{ has a finite subcover from } \mathcal{C}\}.$$

Note that $a \in U$ for some $U \in \mathcal{C}$ because $\mathcal{C}$ is an open cover of $[a, b]$ so that $[a, a] = \{a\} \subset U$ and thus $a \in D$ with the finite subcover being U , moreover, $D \subseteq [a, b]$ so that D is bounded and thus there exists

$$s = \sup D$$

and $s \leq \sup[a, b] = b$ so that $s \in [a, b]$. Hence, there exists an open set G in the cover $\mathcal{C}$ such that $s \in G$, and since G is open, there is some $\delta > 0$ such that $V_\delta(s) \subset G$, i.e., $(s - \delta, s + \delta) \subset G$. But since $s = \sup D$, there must exist $d \in (s - \delta, s] \cap D$, i.e.

$$[a, d] \subset \bigcup_{i=1}^n G_i$$

and given that $[s, s + \delta) \subset G$, we have that

$$[a, d'] \subset G \cup \bigcup_{i=1}^n G_i$$

for $d' \in [s, s + \delta)$ meaning that $d' \in D$ for $d' \in [s, s + \delta) \cap [a, b]$, we have proved that

$$[s, s + \delta) \cap [a, b] \subseteq D \subseteq [a, b]$$

so that taking the supremum we have

$$b \leq s \leq b \iff s = b.$$

Here, we used $\sup([s, s + \delta) \cap [a, b]) = b$, which is true unless $s + \delta \leq b$, but notice that if that were the case, then the supremum would be $s + \delta$ and thus we would get $s + \delta \leq s$ which is equivalent to $\delta \leq 0$, i.e., a contradiction. In particular, given that $s = b$, $b \in [b, b + \delta) \cap [a, b]$ so that $b \in D$ and hence

$$[a, b] \subset G \cup \bigcup_{i=1}^n G_i,$$

so that $[a, b]$ has a finite subcover, as required.

Case 2) Now for any K closed and bounded, let $\mathcal{C}$ be an open cover of K , then since K is bounded, by definition we have that K is bounded below and bounded above, say by α, β , and hence $K \subseteq [\alpha, \beta]$ so that $\mathcal{C} \cup (\mathbb{R} \setminus K)$ is an open cover of $\mathbb{R}$ and hence is an open cover of $[\alpha, \beta]$, but by the previous case we have that there exists a finite subcover of $[\alpha, \beta]$, i.e.,

$$[\alpha, \beta] \subset G_1 \cup G_2 \cup \cdots \cup G_n \cup (\mathbb{R} \setminus K)$$

so that K has a finite subcover.

Theorem 1.6. If $a = b$, then the set has only one element, and thus, by **Theorem 1.1** we have that $m^*([a, b]) = m^*([a]) = 0 = b - a = a - a$. If $a < b$ then take the open cover $(a - \varepsilon/2, b + \varepsilon/2)$ with length $b - a + \varepsilon$ so that $m^*([a, b]) \leq b - a + \varepsilon$ for all $\varepsilon > 0$. We therefore conclude that $m^*([a, b]) \leq b - a$. Conversely, if (I_n) is a countable open cover of $[a, b]$, then by **Theorem 1.4** we have that there exists a finite subcover $\{I_j\}_{j=1}^N$ of (I_n) , i.e.,

$$[a, b] \subset \bigcup_{j=1}^N I_j.$$

We will now show that this implies

$$\sum_{j=1}^N \ell(I_j) \geq b - a.$$

We proceed by induction on N , let $I_j = (a_j, b_j)$. For $N = 1$ we get $[a, b] \subset [a_1, b_1]$ which implies $a_1 \leq a \leq b \leq b_1$ and thus $\ell([a, b]) \leq \ell(I_1)$. Now, suppose the assertion holds true for some $M \in \mathbb{N}$, i.e.,

$$\sum_{i=1}^M \ell(I_i) \geq b - a \quad \text{provided that} \quad [a, b] \subset \bigcup_{i=1}^M I_i$$

and let $\{I_\alpha\}_{\alpha=1}^{M+1}$ be an open cover of $[a, b]$, then b is at least in one of the intervals I_α , which without loss of generality assume that is I_{M+1} so we have $b \in I_{M+1} = (a_{M+1}, b_{M+1})$ which is equivalent to $a_{M+1} < b < b_{M+1}$.

Case 1) Suppose further that $a_{M+1} \leq a$, then $\ell(I_{M+1}) = b_{M+1} - a_{M+1} > b - a_{M+1} \geq b - a$ and therefore we automatically get

$$\sum_{i=1}^M \ell(I_i) \geq \ell(I_{M+1}) > b - a.$$

Case 2) Next, suppose that $a < a_{M+1}$, then we have the inequalities $a < a_{M+1} < b < b_{M+1}$ so that

$$[a, a_{M+1}] \subset \bigcup_{i=1}^M I_i$$

which by the induction hypothesis implies

$$\sum_{i=1}^M \ell(I_i) \geq a_{M+1} - a$$

so that

$$\sum_{i=1}^{M+1} \ell(I_i) \geq a_{M+1} - a + \ell(I_{M+1}) = b_{M+1} - a > b - a.$$

Hence we have that

$$\sum_{n \in \mathbb{N}} \ell(I_n) \geq \sum_{n=1}^M \ell(I_n) > b - a$$

for any countable cover (I_n) of $[a, b]$ and hence, taking the infimum

$$m^*([a, b]) \geq b - a.$$

We therefore conclude that $b - a \leq m^*([a, b]) \leq b - a$, i.e., $m^*([a, b]) = b - a$.

Theorem 1.7. Let $I = [a, b]$ be a (closed) interval. If $a = b$, then $I = \{a\}$ contains only one element, so that if I contains more than one element, then $a < b$ or $a > b$ but the notation $[a, b]$ is reserved for the case $a < b$, so we have that $I = [a, b]$ with $a < b$. Then $m^*(I) = m^*([a, b]) = b - a > 0$. If I were countable, then it has outer measure 0, which is not the case, so we conclude that I is uncountable.

Theorem 1.9. Let $\sim$ be the equivalence relation on $[-1, 1]$ defined by $a \sim b$ if and only if $a - b \in \mathbb{Q}$. Then by the fundamental theorem of equivalence relations we have

$$[-1, 1] = \coprod_{a \in [-1, 1]} [a].$$

Let us now consider a **Vitali set**, i.e., set such that $V \cap [a]$ contains exactly one element for each $a \in [-1, 1]$. Let (r_n) be a sequence of distinct rational numbers such that

$$[-2, 2] \cap \mathbb{Q} = \{r_k\}_{k \in \mathbb{N}},$$

and then if $a \in [-1, 1]$, let v be the unique element in $V \cap [a]$, so that we have $a - v \in \mathbb{Q}$, which implies that $a = r_k + v \in r_k + V$ for some $k \in \mathbb{N}$, i.e.,

$$[-1, 1] \subset \bigcup_{k \in \mathbb{N}} (r_k + V).$$

This implies that

$$2 = m^*([-1, 1]) \leq m^*\left(\bigcup_{k \in \mathbb{N}} (r_k + V)\right) \leq \sum_{k \in \mathbb{N}} m^*(r_k + V) = \sum_{k \in \mathbb{N}} m^*(V),$$

so that $m^*(V) > 0$. Now, suppose that there is some $t \in (r_i + V) \cap (r_j + V)$ for $i \neq j$, then $t = r_i + v = r_j + v$ so that $r_i = r_j$ but since the elements of such sequence were distinct, we get the contradiction $i = j$. Hence the sets $(r_k + V)$ are pairwise disjoint. Now, since $V \subset [-1, 1]$ and $r_k \in [-2, 2]$ for all k , then

$$\bigcup_{k=1}^n (r_k + V) \subset [-3, 3]$$

for any n , and thus

$$m^*\left(\bigcup_{k=1}^n (r_k + V)\right) \leq 6.$$

However,

$$\sum_{k=1}^n m^*(r_k + V) = nm^*(V)$$

and since $m^*(V) > 0$ we can choose n big enough to make $nm^*(V) > 6$ so that the inequality in σ -subadditivity becomes strict, i.e.,

$$m^*\left(\bigcup_{k=1}^n (r_k + V)\right) < \sum_{k=1}^n m^*(r_k + V),$$

where the $(r_k + V)$ are disjoint, i.e., the outer measure m^* is not additive.

Theorem 1.10. Suppose that there exist a function $\mu : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ with

i) $\mu(I) = \ell(I)$ for every open interval I of $\mathbb{R}$.

ii) $\mu\left(\coprod_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} \mu(A_n)$.

iii) $\mu(A + t) = \mu(A)$ for all $A \subset \mathbb{R}$ and every $t \in \mathbb{R}$.

a) μ vanishes on the empty set. Pf: By i) we have $\mu(\emptyset) = 0$

b) μ is monotone. Pf: If $A \subseteq B \subseteq \mathbb{R}$, then we can write $B = (B \setminus A) \cup A$, and thus $\mu(B) = \mu(A) + \mu(B \setminus A)$ so that $\mu(B) \geq \mu(A)$.

- c) μ is a good generalisation on compacts. Pf: For any $a < b$ in $\mathbb{R}$ we have that $(a, b) \subset [a, b] \subset (a - \varepsilon/2, b + \varepsilon/2)$ so that by i) and b) we have $b - a \leq \mu([a, b]) \leq b - a + \varepsilon$ and since $\varepsilon > 0$ was arbitrary, we conclude that $\mu([a, b]) = b - a$.
- d) μ is σ -subadditive. Pf: Consider a sequence (A_n) in $\mathcal{P}(\mathbb{R})$, and then construct the sequence (D_n) defined via

$$D_1 = A_1, \quad D_n = A_n \setminus \bigcup_{i=1}^{n-1} A_i \quad n > 1,$$

notice that D_1 is disjoint from all the other D_n 's and if $n \neq m$ (WLG assume that $n < m$) we have

$$\begin{aligned} D_n \cap D_m &= \left(A_n \setminus \bigcup_{i=1}^{n-1} A_i \right) \cap \left(A_m \setminus \bigcup_{i=1}^{m-1} A_i \right) = \bigcap_{i=1}^{n-1} (A_n \setminus A_i) \cap \bigcap_{i=1}^{m-1} (A_m \setminus A_i) \\ &= \bigcap_{i=1}^{n-1} (A_n \setminus A_i) \cap \bigcap_{i=1}^{n-1} (A_m \setminus A_i) \cap \bigcap_{i=n}^{m-1} (A_m \setminus A_i) \end{aligned}$$

notice that the first term has A_n in each intersection and the last term has $A_m \setminus A_n$ in the first index ($i = n$) so that using associativity and commutativity of $\cap$ we get that there is a term of the form $(A_n \setminus A_1) \cap (A_m \setminus A_n) = \emptyset$ and thus everything is just $\emptyset$. Since (D_n) is a sequence of disjoint subsets, we have that

$$\mu \left(\coprod_{n \in \mathbb{N}} D_n \right) = \sum_{n \in \mathbb{N}} \mu(D_n)$$

but notice how $D_n \subseteq A_n$ for all n so $\mu(D_n) \leq \mu(A_n)$ for all $n \in \mathbb{N}$ by b) and hence

$$\mu \left(\coprod_{n \in \mathbb{N}} D_n \right) \leq \sum_{n \in \mathbb{N}} \mu(A_n)$$

but also

$$\coprod_{n \in \mathbb{N}} D_n = \bigcup_{n \in \mathbb{N}} A_n$$

so the conclusion now readily follows.

We therefore conclude that μ satisfies the same properties of m^* used in the previous proof. This implies that it is not σ -additive (a Vitali set being the counterexample), which is a contradiction.